Unit 2

Mining Association Rules in Large Databases

Association rule mining is used to find patterns, correlations, or relationships among a set of items in large databases. The goal is to discover strong rules using measures like support, confidence, and lift. For example, if people buy bread, they are also likely to buy butter. Such insights help in decision-making and predicting customer behavior.

Association Rule Mining

Association rules are if-then statements that show the relationship between items. For example, Milk → Bread means if milk is bought, bread is likely to be bought too. This is useful in many areas like retail, web usage, and bioinformatics.

It is commonly used in **market basket analysis**, where businesses analyze customer buying habits.

Structure of a Rule:

A rule is written in **"IF → THEN"** format: **A → B**
Which means: *If item A is bought, then item B is likely to be bought.*

Example:

- IF a customer buys **milk**, THEN they also buy **bread**.

To check how good a rule is, we use these three measures:

1. **Support**
    - Tells us how frequently an itemset appears in the database.
    - Example: If 2 out of 10 transactions include both milk and bread, the support = 2/10 = 0.2 or 20%.
2. **Confidence**
    - Tells how often item B is purchased when item A is purchased.
    - Formula: Confidence(A → B) = Support(A and B) / Support(A)
3. **Lift**
    - Tells us how much more likely item B is bought when A is bought, compared to B being bought randomly.
    - Lift > 1 means the rule is useful.

Need Of Association Rule:

- Helps in **product placement** (place commonly bought items close together).
- Used in **recommendation systems** (like Amazon, Netflix).
- Assists in **inventory management** and **target marketing**.

Market Basket Analysis: Association Rule Mining

Market Basket Analysis is the process of analyzing customer purchasing patterns by looking for combinations of items that are bought together.

It is the most common use case of association rule mining.

Steps in Market Basket Analysis:

1. **Prepare the transaction data** – a list of items bought together.
2. **Find frequent itemsets** – groups of items that appear together often.
3. **Generate association rules** – using support and confidence.
4. **Evaluate and use the rules** – to improve sales strategies.

Note:

How Thresholds (Support & Confidence) Are Set in Association Rule Mining

In Association Rule Mining, we use thresholds to filter out only the most useful and reliable rules. These thresholds are set by the data analyst, business expert, or system designer, based on the specific problem, dataset, or business goal.

1. Minimum Support Threshold

- **Definition:** The minimum proportion of transactions that must contain the itemset.
- **Purpose:** Filters out itemsets that occur too rarely to be meaningful.

How to set:

- **Based on dataset size:**
  - If the dataset is small, use a lower support (e.g., 20–30%).
  - For large datasets, use higher support (e.g., 50% or more) to avoid too many rules.
- **Based on domain knowledge:**
  - If a retailer knows that an item like milk is usually bought often, the support threshold can be adjusted accordingly.

2. Minimum Confidence Threshold

- **Definition:** The minimum proportion of times the rule is found to be true.
- **Purpose:** Filters out rules that are not reliable or predictive enough.

How to set:

- **Use domain expertise:**
  - For rules to be useful in practice, confidence should generally be above 70%.
- **Test different values:**
  - You can experiment with values like 0.5, 0.7, or 0.9 and observe how many and what kind of rules are generated.

Rules That Satisfy Thresholds

In Association Rule Mining, we only want to keep the useful and strong rules. For that, we set thresholds:

- Minimum Support = 0.6 (or 60%)
  The rule must be true in at least 60% of all transactions.
- Minimum Confidence = 0.7 (or 70%)
  When the "if" part of the rule happens, the "then" part should also happen at least 70% of the time.

Example:

Given the following transaction data, perform Association Rule Mining and identify the rules that satisfy:

- **Minimum support = 60%**
- **Minimum confidence = 70%**

Transaction Dataset (TID):

Transaction ID	Items Bought
T1	Milk, Bread, Butter
T2	Bread, Butter
T3	Milk, Bread
T4	Milk, Bread, Butter, Eggs
T5	Milk, Bread, Eggs

Step 1: Generate Frequent Itemsets (Support Count)

Let's assume **minimum support = 60%** (i.e., itemset must appear in at least **3 out of 5 transactions**).

Single Items:

- Milk → 4 times
- Bread → 5 times
- Butter → 3 times
- Eggs → 2 times

Milk, Bread, Butter are frequent (support ≥ 3)
Eggs is **infrequent**

Item Pairs (2-itemsets):

- Milk & Bread → 4
- Milk & Butter → 2
- Bread & Butter → 3
- Bread & Eggs → 2
- Milk & Eggs → 2
- Butter & Eggs → 1

Frequent:

- Milk & Bread (4)
- Bread & Butter (3)

Others are infrequent

Step 2: Generate Association Rules from Frequent Itemsets

Let's generate rules from frequent itemset: **{Milk, Bread}**

Possible Rules:

1. **Milk → Bread**
2. **Bread → Milk**

Now calculate **Support** and **Confidence**:

Rule 1: Milk → Bread

- Support(Milk ∩ Bread) = 4/5 = **0.8**
- Confidence = Support(Milk ∩ Bread) / Support(Milk)
  = 4/4 = **1.0 (100%)**

Rule 2: Bread → Milk

- Support = 4/5 = **0.8**
- Confidence = 4/5 = **0.8 (80%)**

Rules that satisfy:

- Minimum **support = 0.6**
- Minimum **confidence = 0.7**

Selected Rules:

1. Milk → Bread (Support: 0.8, Confidence: 1.0)
2. Bread → Milk (Support: 0.8, Confidence: 0.8)

These rules tell us that **Milk and Bread are highly associated**, and customers buying one are very likely to buy the other.

Example:

Consider the following transaction data from a supermarket:

Transaction ID	Items Bought
T1	Milk, Bread, Butter
T2	Bread, Butter
T3	Milk, Bread
T4	Milk, Butter
T5	Bread, Butter

Generate **frequent itemsets** and find **strong association rules** with **minimum support = 60%** and **minimum confidence = 80%**.

Answer:

Step 1: Transaction List

We have 5 transactions:

- T1: Milk, Bread, Butter
- T2: Bread, Butter
- T3: Milk, Bread
- T4: Milk, Butter
- T5: Bread, Butter

Step 2: Find Frequent Itemsets

1-itemsets:

Item	Count	Support (%)
Milk	3	60%
Bread	4	80%
Butter	4	80%

All are frequent (≥ 60%)

2-itemsets:

Itemset	Count	Support (%)
Milk, Bread	2	40%
Milk, Butter	2	40%
Bread, Butter	3	60%

Only **Bread, Butter** is frequent (≥ 60%)

3-itemsets:

Itemset	Count	Support (%)
Milk, Bread, Butter	1	20%

Not frequent (only 20%)

Frequent Itemsets:

- {Milk}, {Bread}, {Butter},
- {Bread, Butter}

Step 3: Generate Association Rules from Frequent Itemsets

From {Bread, Butter} → Possible Rules:

1. **Bread ⇒ Butter**
    - Support = 3/5 = 60%
    - Confidence = 3/4 = 75% (Not ≥ 80%)
2. **Butter ⇒ Bread**
    - Support = 3/5 = 60%
    - Confidence = 3/4 = 75% (Not ≥ 80%)

No rules from 2-itemsets pass both thresholds.

Final Answer:

Frequent Itemsets (Support ≥ 60%):

- {Milk}, {Bread}, {Butter}, {Bread, Butter}

Strong Association Rules (Confidence ≥ 80%):

- None found in this example with current thresholds.

Advanced Metrics: Lift & Conviction

These metrics are used to measure the **strength** and **interestingness** of association rules like:

Milk → Bread

They help determine if the rule is **statistically significant** and **non-random**.

1. Lift:

Lift measures **how much more often** the items in A → B occur **together** than expected if they were **independent**.

$$\text{Lift}(A \to B) = \frac{P(A \cap B)}{P(A) \cdot P(B)} = \frac{Confidence(A \to B)}{Support(B)}$$

Interpretation of Lift Values:

Lift Value	Meaning
= 1	A and B are **independent** (no association)
> 1	A and B are **positively correlated** (stronger than chance)
< 1	A and B are **negatively correlated** (less likely to occur together)

Example:

Assume:

- Support(Milk ∩ Bread) = 0.4
- Support(Milk) = 0.5
- Support(Bread) = 0.6

Then:

$$Confidence(Milk \to Bread) = \frac{0.4}{0.5} = 0.8$$

$$Lift = \frac{0.8}{0.6} = 1.33$$

Interpretation: Customers who buy milk are **33% more likely** to buy bread than average — strong association.

2. Conviction:

Conviction evaluates the **implication direction** and tells how much more often A occurs **without** B **than expected** under independence.

$$\text{Conviction}(A \to B) = \frac{1 - P(B)}{1 - Confidence(A \to B)}$$

Interpretation of Conviction Values:

Conviction Value	Meaning
= 1	A and B occur independently (rule is no better than chance)
> 1	There is a **positive implication** (stronger rule)
→ ∞	Rule has **perfect confidence** (Confidence = 1)

Example:

Given:

- Support(Bread) = 0.6
- Confidence(Milk → Bread) = 0.8

Then:

$$Conviction = \frac{1 - 0.6}{1 - 0.8} = \frac{0.4}{0.2} = 2.0$$

Interpretation: The rule Milk → Bread is **2 times better than chance** in implying Bread when Milk is bought.

Real-World Example (Retail):

Rule: Diaper → Beer

- **Confidence** = 0.7
- **Support(Beer)** = 0.5
- Conviction = $\frac{1-0.5}{1-0.7} = \frac{0.5}{0.3} = 1.67$

Interpretation: This rule is **1.67 times more reliable** than a random guess. Not perfect, but useful!

Association Rule Mining: A Road Map

1. Understanding the Basics:

- **Transactions:** A unit of data containing a collection of items, like a purchase in a market basket analysis example.
- **Itemsets:** A collection of items, like a set of products bought together.
- **Support:** The frequency with which an itemset appears in the dataset.
- **Confidence:** The probability that the consequent (the "then" part of the rule) occurs given the antecedent (the "if" part of the rule).
- **Association Rule:** An "if-then" statement expressing a relationship between itemsets, such as "if a customer buys bread, they are likely to buy butter".

2. The Apriori Algorithm (A Common Approach):

- The Apriori algorithm is a popular method for finding frequent itemsets.
- It works by iteratively identifying frequent subsets of items and then generating rules from those subsets.
- It uses the "Apriori property" that all subsets of a frequent itemset must also be frequent.

3. Steps in the Process:

- **Data Preparation:** Ensure data is in a format suitable for mining, often a transaction database.
- **Frequent Itemset Mining:** Use an algorithm like Apriori to identify itemsets that meet a minimum support threshold.

- **Rule Generation:** Generate association rules from the frequent itemsets, evaluating their confidence.
- **Rule Evaluation and Selection:** Choose rules that are both frequent and have high confidence.
- **Interpretation and Application:** Analyze the discovered rules and use them to make data-driven decisions.

4. Applications of Association Rule Mining:

- **Market Basket Analysis:** Understanding which items are frequently bought together to improve sales and marketing.
- **Fraud Detection:** Identifying patterns in transactions that may indicate fraudulent activity.
- **Customer Behavior Analysis:** Understanding how customers interact with a website or store.
- **Inventory Management:** Optimizing inventory levels by anticipating demand.
- **Cross-Selling and Upselling:** Identifying opportunities to suggest additional products or services.

Mining Multilevel Association Rules from Transaction Databases:

In real-world applications like retail, data items are often organized in a hierarchy or taxonomy (e.g., "Milk" → "Dairy" → "Beverages").

Traditional association rule mining finds rules at a **single concept level**. However, in multilevel association rule mining, we uncover relationships **at multiple abstraction levels.** **OR**

Multilevel Association Rule Mining helps us to find patterns at different levels of abstraction in a data hierarchy.

Example:
Category: Dairy → Milk → Amul Milk
Level: L1 L2 L3

These are association rules derived from data involving items at different levels of a hierarchy.

Example:

- **Level 1 Rule**: Dairy → Bakery
- **Level 2 Rule**: Milk → Bread
- **Level 3 Rule**: Amul Milk → Harvest Bread

Such rules allow more **flexibility** and **detail** in pattern discovery.

Need of Multilevel Rules

1. **Improves Insight**: Higher-level rules give broader trends; lower levels give specifics.
2. **Handles Sparse Data**: Detailed items may appear rarely, but abstracting them into higher categories helps in mining.
3. **Better Decision Making**: Business can act on general patterns or target specific brands.

Approaches for Mining Multilevel Association Rules

Mining association rules at multiple levels of abstraction requires special techniques. There are three main approaches to do this:

1. Top-Down Progressive Deepening Approach

Idea:

- Start from the highest level (like category: Food, Electronics).
- Gradually go deeper into lower levels (like sub-category, brand).
- At each level, find frequent itemsets and generate rules.

Steps:

1. Set a minimum support for each level (lower levels need higher support).
2. Mine frequent itemsets at level 1.
3. Use level 1 results to find level 2 itemsets, and so on.

Example:

- Start with: Food → Snacks
- Then go deeper: Snacks → Chips
- Finally: Chips → Lays

Advantages:

- Saves time by pruning less promising paths early.
- Useful when higher-level patterns are more important.

2. Bottom-Up Approach

Idea:

- Start from the lowest level (most specific, like brand names).
- Move up the hierarchy to more general categories.

Steps:

1. Find frequent itemsets for specific items (e.g., "Lays Chips").
2. Generalize them to next higher levels (e.g., "Chips", then "Snacks").

Example:

- First find: Lays → Pepsi
- Then generalize: Chips → Soft Drinks
- Then further: Snacks → Beverages

Advantages:

- Finds more specific patterns, which may be more actionable.

Disadvantage:

- May miss general patterns that are important for broader strategies.

3. Hybrid Approach

Idea:

- Combines both top-down and bottom-up approaches.
- First identify promising levels using top-down, then go deep using bottom-up.

Steps:

1. Top-down to prune irrelevant paths.
2. Bottom-up to find detailed patterns in selected areas.

Advantages:

- Balanced and more flexible.
- Can adapt to different types of data and goals.

Steps in Mining Multilevel Association Rules

1. **Transform data** to include hierarchical levels.
2. **Set minimum support** for each level.
3. **Generate frequent itemsets** at each level using Apriori or similar algorithms.
4. **Prune uninteresting rules** using confidence, lift, etc.
5. **Generate rules** at all desired levels.

Types of Multilevel Rules:

1. **Intra-level Rules**: Rules at the same level
   → Snacks → Beverages
2. **Inter-level Rules**: Rules across different levels
   → Chips → Coca-Cola
3. **Mixed-level Rules**: Rules involving items at any levels
   → Snacks → Pepsi

Advantages:

- Gives both **overview and detailed** insight.
- Helps companies in **product placement, promotions, and inventory management**.
- Improves accuracy of recommendations and business strategy.

Example:

Given the following transaction data **after hierarchy encoding**, identify **Level-2** and **Level-3** association rules using multilevel association rule mining.

Transactional Data:

Transaction	Items
T1	1** (Milk), 2** (Bread)

Transaction	Items
T2	1A (Amul Milk), 2A (White Bread)
T3	1B (Mother Dairy Milk), 2B (Brown Bread)

Answer:

Objective:

Generate:

- Level 2 Rule: Milk → Bread
- Level 3 Rule: Amul Milk → White Bread

Evaluate each rule using:

- Support
- Confidence

Item Encoding Based on Hierarchy:

Code	Meaning	Level
1**	Milk (Generic)	2
1A	Amul Milk	3
1B	Mother Dairy Milk	3
2**	Bread (Generic)	2
2A	White Bread	3
2B	Brown Bread	3

Step 1: Transform to Level 2 View

At **Level 2**, group all 1A, 1B → 1** (Milk),

and all 2A, 2B → 2** (Bread)

Transaction	Level 2 Items
T1	1**, 2**
T2	1**, 2**
T3	1**, 2**

All 3 transactions contain both Milk and Bread.

Level 2 Rule: Milk → Bread

- Support Count of Milk ∩ Bread = 3
- Support Count of Milk = 3
- Confidence = 3/3 =1.0

Result:

- **Support**: 3/3=1.0
- **Confidence**: 1.0

  Strong Rule

Step 2: Level 3 Analysis (Specific Brands)

Only consider **exact items**, no generalization.

Transaction	Level 3 Items
T2	1A (Amul Milk), 2A (White Bread)
T3	1B, 2B
T1	Skipped (uses only generic codes)

Level 3 Rule: Amul Milk → White Bread

- Support Count of 1A ∩ 2A = 1 (T2)
- Support Count of 1A = 1 (T2)
- Confidence = 1/1=1.0

Result:

- **Support**: 1/3 ≈ 0.33
- **Confidence**: 1.0

  High confidence, but **low support** → Not very frequent.

Rule	Level	Support	Confidence	Interesting?
Milk → Bread	2	1.0	1.0	Very Strong
Amul Milk → White Bread	3	0.33	1.0	Weak Support

Conclusion:

Multilevel mining captures both broad (Level 2) and specific (Level 3) buying patterns.

The rule Milk → Bread is a **general pattern**, applicable to most buyers.

Amul Milk → White Bread is more **targeted**, showing a **brand-level affinity**, useful for personalized marketing.

Mining Distance-Based Association Rules

Distance-based association rules consider not just whether two items appear together, but also **how similar or different** the items are. The term *distance* refers to a **numerical measure of dissimilarity** between data values.

In traditional association rule mining, only the presence or absence of items is checked (binary). But sometimes, data is numerical or continuous, like:

- Customer age

- Purchase amount
- Time of transaction

In such cases, we use **distance-based rules** to find interesting patterns.

Example:

Let's say we want to find rules based on **how far apart two transactions are** in terms of time:

- Rule: If someone buys product A, then within 2 days they also buy product B.

Another example using **price**:

- If a customer buys a product costing ₹500, they are likely to buy another item costing **between ₹450 to ₹550**.

Distance Functions

Common distance measures:

Distance Metric	Formula	Use Case
Euclidean	$d(x, y) = \sqrt{\sum(x_i - y_i)^2}$	Spatial / geometric closeness
Manhattan	(d(x, y) = \sum	x_i - y_i
Hamming	Number of differing bits	Binary vectors
Cosine Similarity	$\frac{x \cdot y}{\|x\|\|y\|}$	Text or document similarity

Types of Distance-Based Rules

a) Numerical Attribute Rules

- Example:

  If Customer Age ≈ 25 → Purchase Amount ≈ ₹2000

b) Spatial Association Rules

- Example:

  If a customer lives within 2 km of a mall → high spending frequency

c) Temporal Distance Rules

- Example:

  If two events occur within 5 minutes → probable causality

Mining Process

Step 1: Data Preprocessing

- Normalize or discretize numeric attributes (e.g., age 20–30).
- Define a **threshold distance** (e.g., ε = 5 units) for "closeness".

Step 2: Generate Distance-Based Itemsets

- Convert transactions into itemsets of "approximate" values.

Example:

Transaction	Age	Amount
T1	24	2000
T2	26	1900
T3	45	5000

Using ε = 5:

- T1 & T2 → Similar Age (~25), Similar Amount (~2000)

Step 3: Apply Association Rule Mining (ARM) with Modified Support/Confidence

- Support: Count of approximate matches (within ε distance)
- Confidence: Same as traditional, but computed over similar-value matches

Application:

Domain	Use Case Example
E-commerce	Customers buying items of similar price or size
Retail Analytics	Products placed close in store bought together
Healthcare	Patients with similar symptoms show similar prescriptions
Sensor Networks	Spatial/temporal event correlation
Location Services	Customers within 1 km of store shop more frequently

Advantages

- Works with **numeric and continuous data**
- Discovers **hidden patterns** missed by binary rules
- Useful in **fuzzy logic** and **approximate reasoning**

EXAMPLE:

Given the following transactional data with numerical attributes:

TID	Age	Purchase Amount
T1	24	₹2000
T2	26	₹1950
T3	30	₹4000
T4	23	₹2100

Assume:

- Distance threshold for age (ε_1) = 5
- Distance threshold for purchase amount (ε_2) = 200

Evaluate the distance-based association rule:

If Age ≈ 25 → Purchase ≈ ₹2000

Compute:

1. Support
2. Confidence
3. Determine whether the rule is strong.

Answer:

Step 1: Define Conditions for Matching

We interpret the rule as:

Find transactions where:

- Age is within ±5 of 25 (i.e., between 20 and 30)
- AND Purchase amount is within ±200 of 2000 (i.e., between ₹1800 and ₹2200)

Step 2: Identify Transactions Matching the Rule

Check each transaction:

TID	Age	In Age Range (20–30)?	Amount	In Amount Range (₹1800–₹2200)?	Match Rule?
T1	24	Yes	2000	Yes	Yes
T2	26	Yes	1950	Yes	Yes
T3	30	Yes (boundary)	4000	No	No
T4	23	Yes	2100	Yes	Yes

Matching Transactions: **T1, T2, T4**

Step 3: Compute Metrics

1. Support

Support = (Number of transactions satisfying both conditions) / (Total transactions)

- Support = 3 / 4 = 0.75

2. Confidence

Confidence = (Transactions satisfying both conditions) / (Transactions where Age ≈ 25)

All four transactions are within Age ≈ 25 range (20–30), but let's confirm:

TID	Age	In Age Range (20–30)?
T1	24	Yes
T2	26	Yes
T3	30	Yes
T4	23	Yes

- Transactions where Age ≈ 25 = **4**
- Transactions where both Age ≈ 25 AND Amount ≈ ₹2000 = **3**

Correction: Earlier assumption was 3/3, but **T3 also matches age but not amount**

Confidence = 3 / 4 = 0.75

Step 4: Final Result

Metric	Value
Support	0.75
Confidence	0.75